

Transport Engineering Group
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The University of Melbourne
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Editor-in-Chief
Transportation Research Part C: Emerging Technologies

Dear Editor-in-Chief,

Please consider the manuscript “GEL-Ped: Instability-aware guarded ensemble learning for short-horizon crowd prediction under flow shift” for publication in *Transportation Research Part C: Emerging Technologies*.

Short-horizon pedestrian forecasts support frequent replanning by mobile robots and automated vehicles, as well as local monitoring in crowded stations and terminals. The paper studies a narrow deployment problem: an interaction model trained on one flow can make a rare unstable correction when the encounter topology changes. GEL-Ped turns an observable property of the deployed predictor—local disagreement among independently initialized graph learners—into a conservative action. The graph ensemble remains fully responsible while its spread lies inside a leave-one-complete-run-out calibration envelope. Only an exceedance moves the forecast toward a bounded behavioural residual model.

The evaluation is organized by complete physical runs and deliberately separates a development-informed crossing stress test from cleaner external evidence. In perpendicular crossing, GEL-Ped improves all thirteen runs relative to the graph mean, with a 2.3% mean RMSE reduction and Holm-adjusted exact $p = 0.0037$. The median gain is smaller, and the manuscript diagnoses the influential unstable run rather than hiding it. A geometry-free coarse boosted model is marginally lower in crossing but 5–8% worse in the four other controlled regimes; GEL-Ped preserves graph accuracy there. On all five ETH/UCY leave-one-scene-out tests, the scene-macro change is negligible and non-significant. The paper therefore makes no universal accuracy claim.

The revision includes a five-member graph ensemble and robust median baseline; coarse, constant-velocity, learned-gate, Mahalanobis, and prospective-geometry controls; all fifteen predeclared Holm-adjusted tests; complete per-run values; collision-event counts; frozen threshold sensitivity; route-axis and future-label-filter audits; and recursive error analysis to 3.2 s. Code, immutable partitions, calibration records, processed complete-run outputs, tests, and figure scripts are available in the frozen referee2-final review snapshot.

The manuscript is original, is not under consideration elsewhere, and all submission statements will be confirmed by the author. Thank you for your consideration.

Sincerely,

Amir Ghorbani
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